



Cambridge IGCSE™

MATHEMATICS

0580/22

Paper 2 (Extended)

October/November 2021

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **7** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao – correct answer only

dep – dependent

FT – follow through after error

isw – ignore subsequent working

oe – or equivalent

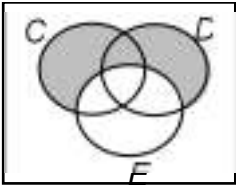
SC – Special Case

nfw – not from wrong working

soi – seen or implied

Question	Answer	Marks	Partial Marks
1	7.5	1	
2	41 43 20	3	B1 for each
3	129	1	
4	79 nfww	3	M2 for $x + x + 58 + 58 + 86 = 360$ oe or $86 - (180 - 2 \times 58)$ implied by $CAB = 22$ or B1 for $DCA = 58$ or $BCA = x$ or $DAC = 64$
5	12	3	M2 for $(95.25 - 15.5) \div 7.25$ oe or $(95.25 - (15.5 - 7.25)) \div 7.25$ oe or M1 for $95.25 - 15.5$ or B1 for 79.75
6	$\frac{1}{3} \times \frac{6}{7}$ oe or $\frac{2}{6} \div \frac{7}{6}$ oe	M1	
	$\frac{2}{7}$ oe	A1	
	their $\frac{2}{7} + \frac{1}{5}$ with a correct method to find fractions with a common denominator	M1	e.g. $\frac{10}{35} + \frac{7}{35}$ oe
	$\frac{17}{35}$ cao	A1	If order of operations not correct SC2 for answer $\frac{10}{41}$ with correct working for $\frac{1}{3} \div \left(\frac{7}{6} + \frac{1}{5} \right)$ or SC1 for $\frac{35}{30} + \frac{6}{30}$ oe
7	$\frac{37}{60}$ oe	4	B3 for $x = 18$ or 37 [yellow] or SC2 for answer $\frac{5}{12}$ or M2 for $\frac{1}{12} = \frac{5}{5+x+2x+1}$ oe or M1 for $5 + x + 2x + 1$ oe or [total number of flowers =] 60

Question	Answer	Marks	Partial Marks
8	2.5 oe	1	
9	$\frac{24}{1000} < 2.1 \times 10^{-1} < 22\% < 0.\dot{2} < \sqrt{0.2}$	2	M1 for four values in the correct order or for conversion to consistent comparable form e.g. 0.21, 0.22..., 0.22, 0.4..., 0.024
10	15	2	M1 for $360 \div (180 - 156)$ or $\frac{180(n-2)}{n} = 156$ oe
11	Straight line from (20, 14) to (35, 14) and straight line from (35, 14) to (45, 0)	3	M1 for $210 \div 14$ soi M1 for $14 \div 1.4$ or any line with gradient -1.4 ending at x axis
12	$13 - 5n$ oe final answer $\frac{n+1}{n}$ oe final answer 2^{n-2} oe final answer	5	B2 for $13 - 5n$ oe final answer or B1 for $-5n + c$ or $13 - kn$ $k \neq 0$ or $13 - 5n$ seen then spoilt B1 for $\frac{n+1}{n}$ oe final answer B2 for 2^{n-2} oe final answer or B1 for 2^{n-k} oe k can be 0
13(a)	3^{6n+5} final answer	2	B1 for 3^5 or $(3^3)^{2n}$ or better or answer $6n + 5$
13(b)	$2^3 \times 3^5 \times p^6$ final answer	2	B1 for two parts correct or $2 \times 3 \times 2 \times 3^2 \times p^3 \times 2 \times 3^2 \times p^3$ or $1944p^6$ or $k^2 = 2^2 \times 3^4 \times p^6$
14(a)	55 Alternate segment theorem	2	B1 for 55
14(b)	Tangents from an external point are equal in length	1	
15(a)	$[y =] 3x + 7$ final answer	3	M1 for $\frac{31-16}{8-3}$ oe M1 for correct substitution of (3, 16) or (8, 31) into $y = (their\ m)x + c$
15(b)	-2	1	

Question	Answer	Marks	Partial Marks																
16(a)	Multiples of 3 <table style="margin-left: auto; margin-right: auto;"> <tr> <td style="padding: 0 10px;">+</td><td style="border: 1px solid black; padding: 5px;">3</td><td style="border: 1px solid black; padding: 5px;">6</td><td style="border: 1px solid black; padding: 5px;">9</td></tr> <tr> <td style="border: 1px solid black; padding: 5px;">2</td><td style="border: 1px solid black; padding: 5px;">5</td><td style="border: 1px solid black; padding: 5px;">8</td><td style="border: 1px solid black; padding: 5px;">11</td></tr> <tr> <td style="border: 1px solid black; padding: 5px;">3</td><td style="border: 1px solid black; padding: 5px;">6</td><td style="border: 1px solid black; padding: 5px;">9</td><td style="border: 1px solid black; padding: 5px;">12</td></tr> <tr> <td style="border: 1px solid black; padding: 5px;">5</td><td style="border: 1px solid black; padding: 5px;">8</td><td style="border: 1px solid black; padding: 5px;">11</td><td style="border: 1px solid black; padding: 5px;">14</td></tr> </table> Prime numbers	+	3	6	9	2	5	8	11	3	6	9	12	5	8	11	14	2	B1 for at least 4 correct entries
+	3	6	9																
2	5	8	11																
3	6	9	12																
5	8	11	14																
16(b)	$\frac{2}{5}$ oe	2	B2FT for $\frac{\text{their } 2}{\text{their } 5}$ or B1FT for $\frac{\text{their } 2}{k}$ k is any integer in the range $1 \leq k \leq 7$ or $\frac{c}{\text{their } 5}$ c is 0, 1 or 2																
17	$\frac{3}{5}$ oe and $-\frac{7}{2}$ oe	1																	
18	$x^2 - 11x + 24 [= 0]$ or $y^2 - 16y + 39 [= 0]$	M2	M1 for $x^2 - 9x + 21 = 2x - 3$ oe or $y = \left(\frac{y+3}{2}\right)^2 - 9\left(\frac{y+3}{2}\right) + 21$ oe																
	$(x - 8)(x - 3) [= 0]$ or $(y - 13)(y - 3) [= 0]$	M1	or for correct factors for <i>their</i> quadratic equation or for correct use of quadratic formula for <i>their</i> equation																
	$[x =] 3$ $[y =] 3$ $[x =] 8$ $[y =] 13$	B2	B1 for one correct pair or two correct x values or two correct y values. If B0 scored and at least 2 method marks scored SC1 for correct substitution of both of <i>their</i> x values or <i>their</i> y values into $y = x^2 - 9x + 21$ or $y = 2x - 3$																
19	 	2	B1 for each																
20(a)	32	2	M1 for $f(6) = 8$ or $ff(x) = 2^{(2^x - 3) - 3}$ oe																

Question	Answer	Marks	Partial Marks
20(b)	$x + 21$	1	
20(c)	-1	2	M1 for $\frac{1}{16}$ oe or 2^{-4} oe
21	$2x^3 - 7x^2 - 12x + 45$ final answer	3	B2 for unsimplified expansion of the three brackets with at most one error or for simplified four-term expression of correct form with three terms correct or B1 for correct expansion of two of the given brackets with at least three terms out of four correct
22	196.6 or 196.60... and 343.4 or 343.39...	3	B2 for one correct angle or M1 for $\sin x = -\frac{2}{7}$ or better If 0 scored SC1 for two angles that sum to 540°
23	$\frac{3y-5}{2(x-12)}$ or $\frac{3y-5}{2x-24}$ final answer	4	SC3 for answer $\frac{3y-5}{x-12}$ or B3 for $(3y-5)(x+12)$ and $2(x-12)(x+12)$ or $(2x-24)(x+12)$ or B2 for $(3y-5)(x+12)$ or $2(x-12)(x+12)$ or $(2x-24)(x+12)$ or $(2x+24)(x-12)$ or B1 for $3y(x+12) - 5(x+12)$ or $x(3y-5) + 12(3y-5)$ or $2(x^2 - 144)$ or $(x-12)(x+12)$